Republic of the Philippines
PARTIDO STATE UNIVERSITY
Camarines Sur

Project Proposal

Title: Establishment of Digital Systems and Instrumentation Laboratory (Phase 1)
College/Campus: College of Engineering and Computational Sciences

I. Introduction

The convergence of education, innovation, and digital technology is reshaping how academic institutions deliver instruction, manage operations, and equip learners with relevant skills. In response to this shift, the College proposes the partial establishment of its Digital Systems and Instrumentation Laboratory (DSIL), targeting at least 30% completion for the initial phase.

The DSIL will serve a dual purpose: first, as a development and hosting environment for key digital systems that enhance college administration; and second, as a learning space that allows Information Technology and Engineering students to engage in hands-on experiences with application development, server deployment, and emerging technologies such as 3D printing.

Through this initiative, the College will not only improve operational efficiency but also provide a dynamic, practice-based laboratory for students to develop competencies in system design, digital infrastructure, and prototyping, aligning with the institution's outcomes-based education (OBE) model and industry standards.

II. Rationale

As education shifts toward digital transformation, institutions must not only digitize operations but also provide students with immersive environments where theory meets practice. The proposed DSIL embodies this vision, bridging the gap between administrative innovation and student experiential learning.

The initial phase of the DSIL focuses on the development of four integrated application systems: Faculty Portfolio Management, Faculty Appointment Scheduling, Student Profile System, and the College Website. These systems will not only modernize administrative functions but will also be student-built, maintained, and enhanced, thus offering real-world software development exposure.

Additionally, the laboratory will feature 3D printing technology to support students in courses requiring physical prototyping, such as electronics, industrial design, civil engineering, and IoT-based innovation projects. The technology will enable them to translate digital models into tangible outputs, enhancing their understanding of the design-to-product cycle.

Deploying these systems on a dedicated on-premise server further immerses students in server configuration, web hosting, and cybersecurity, making the DSIL a cornerstone for applied learning and research. This project aligns with institutional goals, CHED's digital transformation directives, and supports SDG 4 (Quality Education) and SDG 9 (Industry, Innovation, and Infrastructure).

III. Purpose and Objectives

The primary goal of this project is to initiate the development and establishment of a functional Digital Systems and Instrumentation Laboratory, with systems and technologies that support both institutional operations and student learning.

Specific Objectives:

1. **Develop and deploy four digital systems** to support college operations and serve as student-led development projects:
    - Faculty Portfolio Management System
    - Faculty Appointment Scheduling System
    - Student Profile Management System
    - College Website and CMS Portal
2. **Set up a dedicated on-premise server** for hosting the developed applications and training students in server administration and web deployment.
3. **Integrate 3D printing technology** to support digital prototyping and physical model generation for student-led innovation and design courses.
4. **Create a hands-on learning environment** where students can practice system development, infrastructure setup, and maintenance in a controlled and supervised academic setting.
5. **Promote interdisciplinary collaboration** across IT, engineering, and allied programs through joint use of the DSIL facilities and technologies.

IV. Expected Outputs/Deliverables

1. **Four Deployed Application Systems:**
    - **Faculty Portfolio Management System** – to archive faculty accomplishments and documents.
    - **Faculty Appointment System** – for schedule tracking, coordination, and productivity monitoring.
    - **Student Profile System** – for centralized student data storage and academic analytics.
    - **College Website** – including content management features for departments and academic units.
2. **Installed Server Infrastructure:**
    - Physical server with secure internal hosting
    - Backup and recovery systems
    - Network setup for internal access and scalability
3. **3D Printing Setup:**
    - 1 mid-range educational-grade 3D printer
    - Filaments and materials for basic prototyping
    - Safety protocols and training module
4. **Student Learning Materials and Workshops:**
    - Admin and user documentation
    - Development logs and code repositories
    - Workshops on system deployment and 3D modeling
5. **Faculty and Student Engagement:**
    - Opportunities for IT and engineering students to lead system development, test deployment, and prototyping
    - Scheduled usage of 3D printer for instructional and capstone projects

V. Project Budget Breakdown (₱275,000.00)

Item	Estimated Cost (PHP)	Description
A. Hardware, Server, Laboratory Infrastructure		
- Tower Server unit (entry level, 16GB RAM, Intel Xeon E-2324G 4C 65W 3.1GHz)	₱110,000.00	A dedicated physical server to host the four developed systems. The server is equipped with a Xeon processor and 16GB RAM to ensure stable performance and scalability. It provides a controlled environment for student training on server administration, database management, and local hosting.
- FDM (Fused Deposition Modeling) 3D printer	₱59,000.00	An educational-grade 3D printer that allows students to prototype models, parts, and system enclosures. It is particularly useful in IT, engineering, and design-related disciplines to facilitate project-based learning, particularly in the areas of IoT, robotics, and digital fabrication.
- Hyper PLA RFID 3D Printer Filament 1.75mm 1KG	₱1,040.00 (10pcs) ₱10,400.00	PLA filament rolls (10 kg total) for continuous 3D printing use in classes, prototyping activities, and student research. The RFID version ensures compatibility with smart printers and material tracking.
- TV Monitor 55"	₱23,300.00	A large display screen for demonstrations, workshops, and system monitoring during training and presentations. It enables clear visibility of code walkthroughs, 3D modeling previews, and server dashboards for instructional use.
Subtotal A	₱202,700.00	
B. Laboratory Equipment		
- Split-type 1 HP AC	₱30,000.00	Provides temperature control to ensure optimal operating conditions for server equipment, computers, and 3D printer, especially in enclosed lab environments. Helps maintain hardware longevity and comfort during lab use.
- 2KVA ON-LINE Double Conversion UPS 2000VA/2000W	₱33,500.00	A high-capacity Uninterruptible Power Supply to protect the server and 3D printer from power interruptions. On-line double conversion type ensures constant clean power delivery to sensitive electronics.
- AVR Automatic Voltage Regulator 2kVA SVC-2000	₱6,800.00	A voltage stabilizer protects electronic equipment from voltage fluctuations. This stabilizer guarantees a steady supply of power to delicate systems, particularly in regions susceptible to power surges or drops.
- Wheel 15-Meter Cord Length 10A	₱2,000.00	This industrial-grade extension cord is mounted on a wheel, allowing for a flexible lab layout. It

		facilitates the effortless relocation of powered equipment and ensures safe cable management during training, events, or reconfiguration.
Subtotal B	₱72,300.00	
TOTAL PROJECT COST	**₱275,000.00**	

The establishment of 30% of the Digital Systems and Instrumentation Laboratory marks a pivotal step in the College's transition toward a digitally empowered and learning-centered institution. Beyond modernizing administrative processes, this initiative creates a space where students can actively learn, apply, and innovate using real-world technologies, from server systems to 3D printing.

With an interdisciplinary and outcomes-based orientation, the DSIL is not just an infrastructure upgrade; it is an investment in the future of student capability, research, and institutional excellence.

Source of funds: College Strategic Fund Budget.

Prepared by:

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Certified as to Availability of Funds/Applicability of Object of Expenditures: Internally Generated Fund

AMALIA O. OCAMPO
Supervising Administrative Officer—Finance
Acting Head Budget Section
Date: 1/29/26

NANCY C. ENCINAS
Accountant III
Accounting Section Chief
Date: 1/29/26

Reviewed by:

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Chief Administrative Officer
Date: 1/30/26
Recommending Approval:

Approved:

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Date: _____